



Human–AI Authority Allocation System Using Fuzzy Logic
Fuzzy Logic and Inference Systems (IMAT3722)
Fuzzy Coursework Report

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Abstract

The increasing integration of Artificial Intelligence (AI) into safety-critical and decision-intensive environments has created a growing need for effective mechanisms that determine the appropriate balance of authority between human operators and intelligent systems. While AI systems are capable of processing large volumes of information and making rapid decisions, human operators continue to provide contextual awareness, ethical judgement, and adaptability in uncertain situations. Determining when decision-making authority should be allocated to a human, shared between both entities, or delegated to an AI system remains a significant challenge.

This project presents the design, implementation, testing, and evaluation of a Mamdani Fuzzy Inference System (FIS) developed using MATLAB for Human–AI Authority Allocation. The proposed system utilises three input variables: Human Readiness, AI Readiness, and Mission Severity. Based on these inputs, the system generates a single output variable known as Authority Allocation, representing the level of control assigned to either the human operator, a shared control arrangement, or the AI system.

Fuzzy logic was selected as the underlying decision-making approach due to its ability to model uncertainty and represent expert knowledge through linguistic variables and rule-based reasoning. The developed system employs twenty-seven fuzzy rules, trapezoidal and triangular membership functions, and centroid defuzzification to generate interpretable authority allocation decisions.

A comprehensive testing framework consisting of thirty scenarios, including operational cases, boundary conditions, and additional evaluation cases, was developed to assess system performance. Results demonstrate that the proposed model successfully adapts authority allocation according to changing operational conditions while maintaining transparency and interpretability. The system achieved meaningful differentiation between Human-Led, Shared-Control, and AI-Led decisions, demonstrating the suitability of fuzzy logic for human-centred decision support applications.

The findings indicate that fuzzy inference systems provide an effective framework for modelling collaborative Human–AI decision-making and offer a practical foundation for future research in adaptive autonomy and intelligent control systems.

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System Overview

1.1 Introduction

Artificial Intelligence has become increasingly integrated into modern operational environments including healthcare, transportation, defence, industrial automation, and autonomous systems. Advances in machine learning, computer vision, and intelligent decision support have enabled AI systems to perform tasks that traditionally required significant human involvement. Despite these advancements, complete automation remains unsuitable in many situations due to the need for human judgement, ethical reasoning, situational awareness, and accountability (Russell and Norvig, 2021).

Human–AI collaboration has therefore emerged as an important area of research. Rather than replacing human operators, modern intelligent systems increasingly operate as collaborative partners that support decision-making and task execution. In such environments, determining the appropriate allocation of authority between humans and AI systems becomes critical. Excessive reliance on AI may introduce risks when models encounter unfamiliar situations, while excessive human control may reduce operational efficiency and limit the benefits offered by automation.

Authority allocation refers to the process of determining which decision-making entity should possess control under a given set of operational circumstances. Traditional approaches often rely on rigid threshold-based methods or manually defined policies that struggle to account for uncertainty and ambiguity. Real-world environments rarely consist of clearly defined conditions, making flexible decision-making mechanisms necessary.

Fuzzy logic provides an attractive solution to this challenge. Introduced by Zadeh (1965), fuzzy logic allows reasoning using linguistic concepts such as Low, Medium, and High rather than strict numerical boundaries. This capability enables systems to model human reasoning and handle uncertainty more effectively than conventional binary logic approaches.

The purpose of this project is to develop a fuzzy logic-based Human–AI Authority Allocation System capable of dynamically assigning decision authority based on three operational factors: Human Readiness, AI Readiness, and Mission Severity. The resulting system provides a transparent and interpretable framework for determining whether control should remain with a human operator, be shared collaboratively, or be delegated primarily to an AI system.

The project was implemented using MATLAB's Fuzzy Logic Toolbox and employs a Mamdani Fuzzy Inference System due to its interpretability and widespread adoption in decision-support applications. The developed model demonstrates how fuzzy reasoning can support authority allocation decisions in uncertain environments while maintaining explainability and ease of implementation.

1.2 Problem Statement

As AI systems become increasingly capable, organisations face the challenge of determining how authority should be distributed between human operators and intelligent systems. Existing approaches frequently rely on static policies or fixed decision thresholds that fail to adapt to changing operational conditions.

Consider a scenario in which an AI system demonstrates high readiness and reliability while a human operator is experiencing reduced situational awareness. In such circumstances, greater

authority may reasonably be assigned to the AI system. Conversely, situations involving high mission criticality or uncertain environmental conditions may require increased human oversight regardless of AI capability. Determining the appropriate balance between human judgement and automated decision-making therefore becomes a complex problem involving uncertainty and multiple interacting factors.

Traditional decision models often struggle to capture these nuanced relationships because they require precise boundaries between conditions. Real-world situations, however, frequently contain ambiguity where variables cannot be accurately classified as simply high or low. Consequently, a more flexible approach is required to represent gradual transitions between decision states.

This project addresses this challenge through the development of a fuzzy inference system capable of modelling expert reasoning and generating authority allocation decisions based on multiple inputs. By incorporating fuzzy logic principles, the proposed system aims to provide a transparent and adaptable framework for Human–AI collaboration.

1.3 Aim and Objectives

Aim

To design, implement, test, and evaluate a fuzzy logic-based Human–AI Authority Allocation System capable of dynamically determining appropriate levels of decision-making authority using Human Readiness, AI Readiness, and Mission Severity as input variables.

Objectives

The objectives of this project are as follows:

1. To investigate fuzzy logic principles and their application in intelligent decision-support systems.
2. To design a Mamdani Fuzzy Inference System comprising three input variables and one output variable.
3. To develop suitable fuzzy membership functions representing Human Readiness, AI Readiness, Mission Severity, and Authority Allocation.
4. To construct a comprehensive fuzzy rule base representing expert-inspired authority allocation decisions.
5. To implement the proposed system using MATLAB and the Fuzzy Logic Toolbox.
6. To generate a structured set of test scenarios including operational cases and boundary conditions.
7. To evaluate the behaviour and performance of the system using quantitative and qualitative analysis.
8. To assess the effectiveness of fuzzy logic as a mechanism for Human–AI authority allocation and collaborative decision-making.

1.4 System Inputs and Outputs

The proposed Human–AI Authority Allocation System was designed using three input variables and one output variable. The selected variables represent key factors that influence the distribution of decision-making authority between human operators and intelligent systems within collaborative environments.

Human Readiness represents the capability of a human operator to perform a task effectively. This variable reflects factors such as experience, situational awareness, training, fatigue, workload, and overall preparedness for decision-making. Human Readiness is represented on a scale ranging from 0 to 100 and is divided into three linguistic categories: Low, Medium, and High.

AI Readiness represents the operational capability of the artificial intelligence system. This variable reflects characteristics such as confidence, reliability, accuracy, and the ability of the AI system to successfully complete a task. Similar to Human Readiness, AI Readiness is represented using three linguistic categories: Low, Medium, and High across a numerical range of 0 to 100.

Mission Severity represents the criticality of the operational environment. Situations involving greater risk, uncertainty, or potential consequences are assigned higher severity values. This variable allows the system to consider environmental importance when allocating authority. Mission Severity is also represented using Low, Medium, and High linguistic categories.

The system produces a single output variable known as Authority Allocation. The output represents the degree of decision-making authority assigned within the Human–AI partnership. Lower values indicate that authority should remain primarily with the human operator, while higher values indicate greater delegation to the AI system. Intermediate values represent shared decision-making arrangements. The output is categorised into Human-Led, Shared Control, and AI-Led regions.

Table 1.1 Summary of Input and Output Variables

Variable	Type	Range	Linguistic Categories
Human Readiness	Input	0–100	Low, Medium, High
AI Readiness	Input	0–100	Low, Medium, High
Mission Severity	Input	0–100	Low, Medium, High
Authority Allocation	Output	0–100	Human-Led, Shared Control, AI-Led

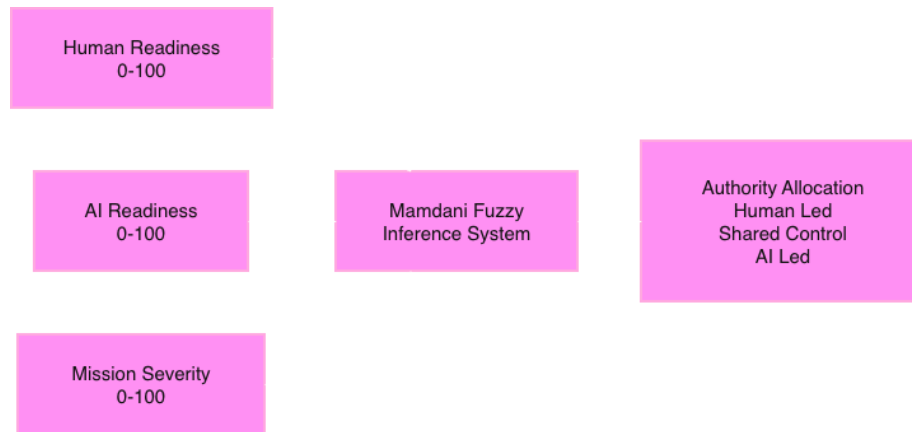


Figure 1.1: Conceptual architecture of the proposed Human–AI Authority Allocation System. Human Readiness, AI Readiness and Mission Severity are processed by a Mamdani Fuzzy Inference System to generate an Authority Allocation recommendation.

1.5 Report Structure

This report is organised into six chapters, each addressing a specific stage of the system development lifecycle.

Chapter 1 introduces the project, presents the motivation behind Human–AI authority allocation, defines the problem statement, and outlines the objectives of the proposed system.

Chapter 2 reviews existing literature related to fuzzy logic, intelligent decision support systems, Human–AI collaboration, authority allocation frameworks, and Mamdani fuzzy inference systems. The review establishes the theoretical foundation that motivates the adoption of fuzzy logic for authority allocation.

Chapter 3 presents the design of the proposed fuzzy inference system. The chapter describes the selection of input and output variables, membership function development, fuzzy rule construction, inference mechanisms, and defuzzification strategies.

Chapter 4 discusses the implementation of the system using MATLAB and the Fuzzy Logic Toolbox. Relevant code segments are presented and explained to demonstrate how the theoretical design was translated into a functioning software system.

Chapter 5 presents the testing methodology and evaluation results. Membership function visualisations, scenario testing results, performance metrics, and authority allocation distributions are analysed to assess system effectiveness.

Chapter 6 concludes the report by summarising key findings, discussing limitations, and proposing opportunities for future development of Human–AI authority allocation systems.

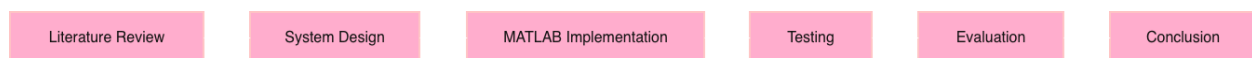


Figure 1.2: Overall project workflow illustrating progression from literature review through system design, implementation, testing, evaluation and final conclusions.

Literature Review

2.1 Introduction

The increasing adoption of Artificial Intelligence (AI) across various industries has created a growing need for effective decision-support mechanisms capable of managing uncertainty and facilitating collaboration between humans and intelligent systems. Traditional decision-making approaches frequently rely on deterministic logic, where conditions are represented using precise boundaries and binary classifications. However, real-world environments often involve ambiguity, incomplete information, and subjective judgement, making deterministic approaches insufficient for many complex decision problems.

Fuzzy logic has emerged as a powerful methodology for addressing such challenges. Since its introduction by Zadeh (1965), fuzzy logic has been widely applied to control systems, decision support, automation, medical diagnosis, risk assessment, and intelligent transportation systems. Unlike conventional binary logic, fuzzy logic allows reasoning using linguistic concepts such as “low”, “medium”, and “high”, thereby enabling systems to model human thinking more naturally.

The objective of this literature review is to examine the theoretical foundations of fuzzy logic, review its application within intelligent decision-support systems, explore research related to Human–AI collaboration, and identify existing approaches to authority allocation. The review establishes the academic basis for the development of the Human–AI Authority Allocation System proposed in this project.

2.2 Foundations of Fuzzy Logic

Fuzzy logic was first introduced by Zadeh (1965) as an extension of classical set theory. Traditional set theory assumes that an element either belongs to a set or does not belong to a set. Membership is therefore represented using binary values of 0 or 1. While this approach is suitable for precise mathematical problems, it struggles to represent many real-world concepts that are inherently vague or uncertain.

For example, when describing an individual's readiness level, it may be difficult to determine whether a readiness score of 69 should be classified as “medium” or “high”. Human reasoning naturally accommodates such ambiguity by allowing partial membership between categories. Fuzzy logic captures this behaviour through the use of membership functions, which assign values between 0 and 1 to represent the degree of membership within a fuzzy set (Ross, 2010).

The fundamental components of a fuzzy inference system include fuzzification, a rule base, an inference engine, aggregation mechanisms, and defuzzification procedures (Mendel, 1995). During fuzzification, crisp numerical inputs are transformed into fuzzy membership values. The

inference engine then evaluates predefined rules that represent expert knowledge. Finally, the aggregated fuzzy outputs are converted back into a crisp numerical value through defuzzification.

The ability of fuzzy systems to model uncertainty while maintaining interpretability has contributed significantly to their adoption in decision-support applications (Zimmermann, 2010). Unlike many machine learning techniques that operate as black-box models, fuzzy systems provide transparent reasoning processes that can be directly examined and understood by human users.

2.3 Evolution of Intelligent Decision-Support Systems

Decision-support systems have evolved considerably over the past several decades. Early systems relied primarily on rule-based expert systems that encoded human expertise using manually defined rules. These systems were particularly popular during the 1970s and 1980s due to their interpretability and ease of implementation (Turban, Sharda and Delen, 2011).

Although expert systems demonstrated success in several domains, their performance was often limited by rigid rule structures and an inability to adapt effectively to uncertainty. As computational capabilities improved, statistical methods and machine learning algorithms became increasingly popular alternatives.

Machine learning approaches introduced the ability to identify hidden relationships within data and automatically generate predictive models. Techniques such as Artificial Neural Networks, Support Vector Machines, Decision Trees, and Random Forests have demonstrated strong predictive performance across numerous applications (Goodfellow, Bengio and Courville, 2016).

Despite these advantages, many machine learning systems suffer from limited interpretability. Complex models frequently operate as black boxes, making it difficult for users to understand the reasoning behind generated decisions. This limitation becomes particularly significant within safety-critical environments where accountability and transparency are essential.

Fuzzy logic occupies a unique position within intelligent decision-support systems because it combines computational reasoning with human interpretability. Fuzzy systems provide clear rule-based explanations while simultaneously accommodating uncertainty and gradual transitions between decision states. Consequently, fuzzy inference systems continue to be widely adopted within domains requiring explainable decision-making (Ross, 2010).

2.4 Human–AI Collaboration

Human–AI collaboration refers to the cooperative interaction between human operators and artificial intelligence systems to achieve shared objectives. Rather than replacing humans entirely, modern intelligent systems increasingly function as collaborative partners that augment human capabilities and support decision-making processes (Shneiderman, 2022).

Research has shown that effective Human–AI collaboration can improve performance, increase efficiency, and reduce cognitive workload when properly designed (Endsley, 2017). However, achieving successful collaboration requires careful consideration of trust, transparency, accountability, and authority distribution.

One of the primary challenges within Human–AI systems is determining the appropriate level of autonomy assigned to the AI component. Excessive automation may result in over-reliance on AI recommendations, while insufficient automation may prevent organisations from fully benefiting from technological capabilities (Parasuraman, Sheridan and Wickens, 2000).

Trust plays a particularly important role within Human–AI partnerships. Operators are more likely to accept AI recommendations when systems provide transparent reasoning and understandable explanations. Conversely, opaque systems may generate resistance or distrust, even when their decisions are technically correct (Lee and See, 2004).

These considerations have motivated significant research into adaptive autonomy and dynamic authority allocation frameworks. Such approaches seek to continuously adjust the balance of control between humans and AI systems according to operational conditions, task complexity, and environmental uncertainty.

2.5 Authority Allocation in Autonomous Systems

Authority allocation refers to the distribution of decision-making responsibility between human operators and automated systems. As autonomous technologies become increasingly capable, determining the appropriate balance between human control and machine autonomy has become an important research challenge.

Early automation systems typically employed fixed levels of autonomy, where either the human operator or the automated system maintained control throughout task execution. Although such approaches simplified system design, they often failed to account for changing operational conditions and varying levels of system performance (Parasuraman, Sheridan and Wickens, 2000).

To address these limitations, researchers proposed adaptive automation frameworks that dynamically adjust authority allocation according to workload, situational awareness, environmental complexity, and system confidence. Adaptive automation seeks to maintain optimal performance by assigning authority to the most capable decision-maker under current conditions (Kaber and Endsley, 2004).

Human factors research has demonstrated that inappropriate authority allocation can negatively impact system performance. Excessive automation may lead to automation bias, reduced situational awareness, and over-reliance on machine recommendations. Conversely, insufficient automation may increase workload and reduce operational efficiency (Endsley, 2017).

Modern Human–AI systems therefore require decision-making mechanisms capable of evaluating multiple factors simultaneously when determining authority allocation. Such mechanisms must remain transparent, interpretable, and capable of operating under uncertainty. Fuzzy logic is particularly well suited to this requirement because it enables authority decisions

to be expressed through human-readable rules while accommodating gradual transitions between decision states.

The authority allocation problem addressed within this project follows the same principles. Rather than assigning authority using fixed thresholds, the proposed system evaluates Human Readiness, AI Readiness, and Mission Severity simultaneously to generate adaptive authority allocation recommendations.

2.6 Mamdani Fuzzy Inference Systems

Among the various fuzzy inference approaches proposed within the literature, the Mamdani Fuzzy Inference System remains one of the most widely adopted due to its simplicity, transparency, and interpretability (Mamdani and Assilian, 1975).

The Mamdani approach was originally developed for controlling a steam engine and boiler system. Since its introduction, it has been successfully applied across a wide range of domains including industrial automation, medical diagnosis, risk assessment, transportation systems, and intelligent decision-support applications (Ross, 2010).

A Mamdani Fuzzy Inference System consists of four primary stages:

1. Fuzzification
2. Rule Evaluation
3. Aggregation
4. Defuzzification

During fuzzification, crisp numerical inputs are transformed into fuzzy membership values. For example, a Human Readiness value of 65 may simultaneously belong to both the Medium and High fuzzy sets with different degrees of membership.

The rule evaluation stage applies expert-defined IF–THEN rules to determine how input conditions influence system outputs. These rules form the knowledge base of the fuzzy system and represent the reasoning process used by domain experts.

Aggregation combines the outputs generated by multiple activated rules into a single fuzzy output set. Finally, defuzzification converts the aggregated fuzzy output into a crisp numerical value that can be interpreted and acted upon by decision-makers.

The transparency of the Mamdani approach provides a significant advantage compared with many machine learning techniques. Each decision can be traced directly to the underlying fuzzy

rules, enabling users to understand how outputs are generated. This characteristic is particularly important in Human–AI systems where explainability and trust are essential (Lee and See, 2004).

Consequently, the Mamdani approach was selected for implementation within this project due to its interpretability, widespread acceptance, and suitability for modelling authority allocation decisions.

2.7 Applications of Fuzzy Logic in Decision-Support Systems

Fuzzy logic has been successfully applied to numerous decision-support applications involving uncertainty and incomplete information. One of the primary reasons for its popularity is its ability to combine quantitative measurements with qualitative expert knowledge.

In healthcare, fuzzy inference systems have been used to support medical diagnosis, patient monitoring, disease prediction, and treatment selection. These applications often involve uncertain symptoms and subjective clinical assessments, making fuzzy reasoning particularly valuable (Kosko, 1994).

Within transportation systems, fuzzy logic has been applied to traffic management, autonomous vehicle control, route optimisation, and collision avoidance systems. The ability to evaluate multiple environmental factors simultaneously allows fuzzy systems to respond effectively to changing operational conditions (Ross, 2010).

Industrial automation represents another major application area. Fuzzy controllers have been widely deployed in manufacturing processes, robotics, and process control systems due to their robustness and ability to operate without precise mathematical models (Mendel, 1995).

Researchers have also explored fuzzy logic within military and defence decision-support systems, where uncertainty, incomplete information, and rapidly changing environments frequently occur. Such systems often require decisions to be made using approximate reasoning rather than exact calculations.

More recently, fuzzy logic has been incorporated into Human–AI collaboration frameworks to support task allocation, trust modelling, and adaptive autonomy. These applications demonstrate the suitability of fuzzy reasoning for problems involving both human and technological factors (Hancock et al., 2011).

The widespread adoption of fuzzy systems across diverse domains provides strong evidence that fuzzy logic represents a suitable methodology for authority allocation problems. Its ability to balance interpretability with uncertainty handling aligns closely with the requirements of Human–AI collaborative environments.

2.8 Research Gap

The literature demonstrates significant progress in fuzzy logic systems, intelligent decision-support frameworks, and Human–AI collaboration models. However, several limitations remain within existing approaches.

Many authority allocation frameworks rely on fixed levels of automation that do not adapt dynamically to changing operational conditions. Such approaches may fail to account for fluctuations in human performance, AI capability, and environmental risk factors (Parasuraman, Sheridan and Wickens, 2000).

Although machine learning techniques have shown strong predictive performance, many operate as black-box models that provide limited transparency and explainability. This limitation is particularly problematic within Human–AI systems where users must understand and trust system recommendations before acting upon them (Shneiderman, 2022).

Furthermore, existing studies frequently focus on individual variables such as workload, trust, or automation level in isolation. Relatively few approaches simultaneously consider human capability, AI capability, and mission criticality within a single interpretable decision-making framework.

Consequently, there remains a need for a transparent authority allocation model capable of integrating multiple operational factors while maintaining explainability and ease of implementation. This project addresses that gap through the development of a Mamdani Fuzzy Inference System that combines Human Readiness, AI Readiness, and Mission Severity to generate adaptive authority allocation recommendations.

By employing fuzzy membership functions and expert-inspired rule-based reasoning, the proposed system seeks to provide a practical and interpretable solution for Human–AI authority allocation problems.

2.9 Chapter Summary

This chapter reviewed the theoretical foundations underpinning the proposed Human–AI Authority Allocation System. The discussion began with an examination of fuzzy logic principles and their advantages for modelling uncertainty. The evolution of intelligent decision-support systems and Human–AI collaboration frameworks was then explored, highlighting the growing importance of adaptive authority allocation mechanisms.

The literature demonstrated that Mamdani Fuzzy Inference Systems provide a transparent and interpretable approach for decision-making under uncertainty. Previous research also showed that fuzzy logic has been successfully applied across healthcare, transportation, industrial automation, and intelligent control applications.

A research gap was identified concerning the limited availability of interpretable authority allocation frameworks capable of simultaneously considering human readiness, AI readiness, and mission severity. The proposed system addresses this gap through the development of a fuzzy logic-based decision-support framework implemented using MATLAB.

The next chapter presents the design of the proposed Human–AI Authority Allocation System, including system architecture, membership function development, rule base construction, and fuzzy inference mechanisms.

System Design

3.1 Introduction

This chapter presents the design of the proposed Human–AI Authority Allocation System. The system was developed using a Mamdani Fuzzy Inference System (FIS) to determine the appropriate distribution of decision-making authority between human operators and artificial intelligence systems.

The design process focused on developing a transparent and interpretable decision-support framework capable of handling uncertainty within Human–AI collaborative environments. Three input variables were selected to represent the operational state of both decision-making entities and the surrounding environment. These inputs are processed through a fuzzy rule base to generate an authority allocation recommendation.

The chapter describes the overall system architecture, the rationale behind variable selection, the development of membership functions, the construction of the rule base, and the fuzzy inference process used to generate system outputs.

3.2 System Architecture

The proposed Human–AI Authority Allocation System follows a conventional Mamdani fuzzy inference architecture. The system accepts three operational inputs and generates a single authority allocation output.

The three inputs are:

1. Human Readiness
2. AI Readiness
3. Mission Severity

These inputs are processed through a fuzzy inference engine containing a knowledge base composed of twenty-seven fuzzy rules. The resulting fuzzy outputs are aggregated and converted into a crisp authority allocation score through centroid defuzzification.

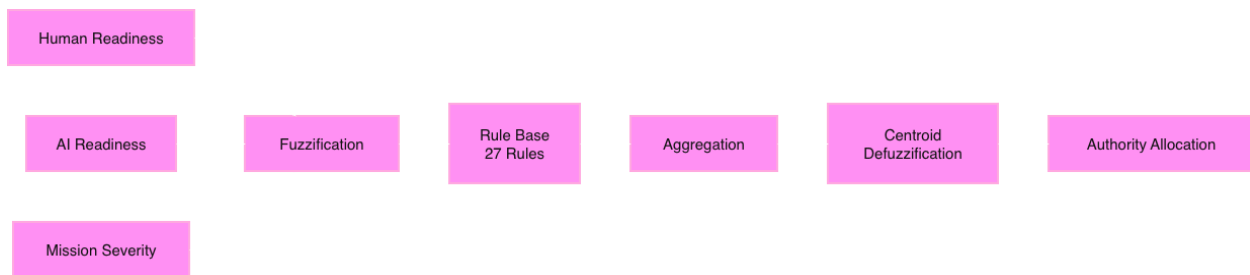


Figure 3.1: Internal architecture of the Mamdani Fuzzy Inference System showing fuzzification, rule evaluation, aggregation and defuzzification stages.

As illustrated in Figure 3.1, Human Readiness, AI Readiness, and Mission Severity first undergo fuzzification. The fuzzified values are then evaluated against the rule base. Following rule evaluation and aggregation, the resulting fuzzy output is converted into a crisp authority allocation value. This value determines whether decision-making authority should remain primarily with the human operator, be shared between both entities, or be delegated to the AI system.

The architecture was intentionally designed to remain simple, interpretable, and computationally efficient while still capturing the uncertainty inherent within Human–AI decision environments.

3.3 Selection of Input Variables

The effectiveness of any fuzzy inference system depends heavily on the selection of appropriate input variables. The variables chosen for this project were selected because they directly influence the allocation of authority within collaborative Human–AI systems.

3.3.1 Human Readiness

Human Readiness represents the capability of a human operator to effectively participate in decision-making. This variable reflects factors such as experience, training, situational awareness, cognitive workload, and overall preparedness.

A highly prepared operator is generally capable of maintaining greater decision authority, whereas reduced readiness may justify increased support from automated systems. Consequently, Human Readiness was selected as a critical input variable within the authority allocation process.

The variable is represented on a numerical scale ranging from 0 to 100 and divided into three linguistic categories:

- Low
- Medium
- High

3.3.2 AI Readiness

AI Readiness represents the capability of the artificial intelligence system to perform the assigned task successfully. Factors influencing AI readiness may include model confidence, reliability, sensor quality, environmental conditions, and system performance.

High AI readiness indicates that the automated system can reliably support decision-making and potentially assume greater authority. Conversely, lower readiness suggests that greater human oversight may be required.

The variable is represented using the same numerical range and linguistic categories as Human Readiness.

3.3.3 Mission Severity

Mission Severity represents the level of operational risk or criticality associated with a particular situation. High-severity environments often require greater caution due to the potential consequences of incorrect decisions.

Including Mission Severity ensures that authority allocation decisions consider not only the capabilities of the human and AI entities but also the significance of the surrounding operational context.

The variable is represented using three linguistic categories:

- Low
- Medium
- High

Table 3.1 Description of Input Variables

Variable	Description	Range
Human Readiness	Human operator preparedness and capability	0–100
AI Readiness	AI system reliability and capability	0–100
Mission Severity	Operational risk and criticality	0–100

3.4 Membership Function Design

Membership functions form the foundation of fuzzy reasoning by defining how numerical inputs are translated into linguistic concepts. The selection of membership functions directly influences system behaviour and output generation.

The proposed system employs a combination of trapezoidal and triangular membership functions. Trapezoidal functions were selected for the Low and High categories because they provide stable representation of extreme conditions. Triangular membership functions were selected for Medium categories because they offer smooth transitions between neighbouring fuzzy sets.

This combination provides a balance between interpretability and computational simplicity.

3.4.1 Human Readiness Membership Functions

Human Readiness was represented using three fuzzy sets:

- Low
- Medium
- High

The parameters were selected to provide gradual overlap between categories and prevent abrupt classification changes.

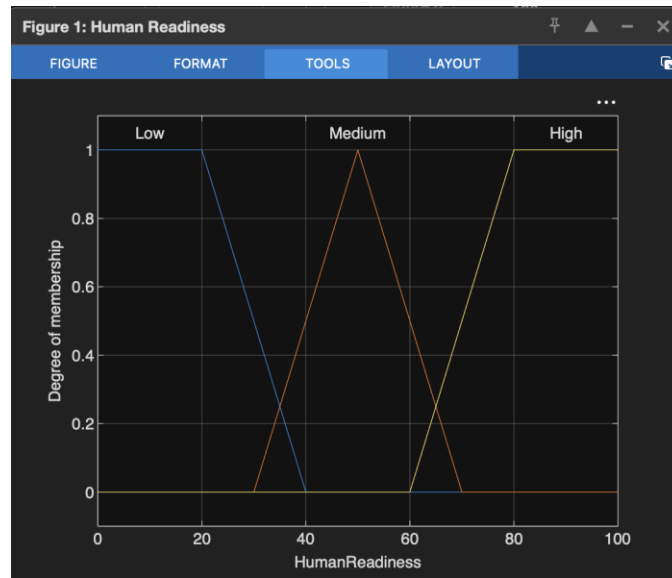


Figure 3.2: Human Readiness membership functions illustrating overlap between Low, Medium and High readiness states.

The overlap between adjacent membership functions ensures smooth transitions as readiness levels change.

3.4.2 AI Readiness Membership Functions

AI Readiness uses the same fuzzy structure as Human Readiness to maintain consistency across capability-related variables.

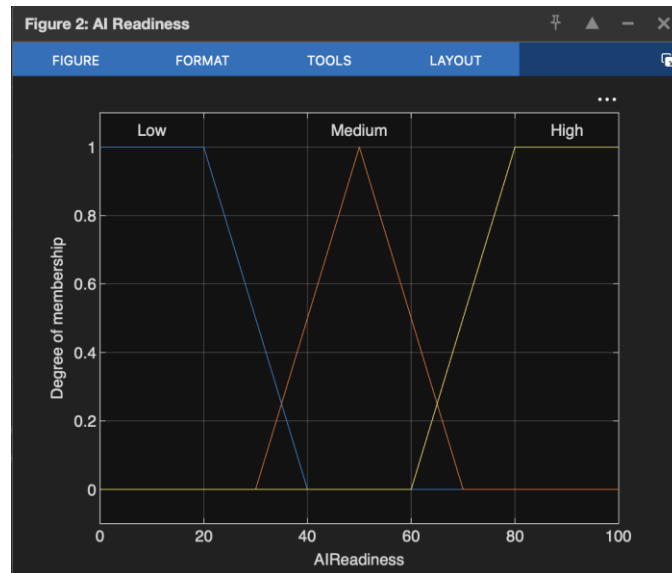


Figure 3.3: AI Readiness membership functions showing gradual transitions between capability levels.

3.4.3 Mission Severity Membership Functions

Mission Severity was represented using three fuzzy categories corresponding to low-risk, moderate-risk, and high-risk operational conditions.

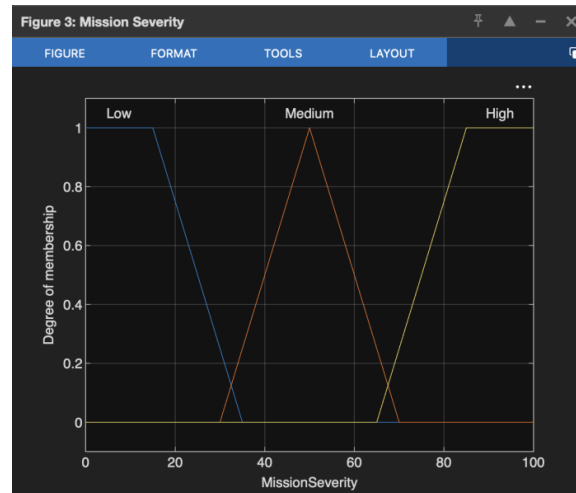


Figure 3.4: Mission Severity membership functions representing low, medium and high operational criticality.

3.4.4 Authority Allocation Membership Functions

The output variable Authority Allocation was divided into three linguistic categories:

- Human-Led
- Shared Control
- AI-Led

These categories represent increasing levels of authority delegation toward the AI system.

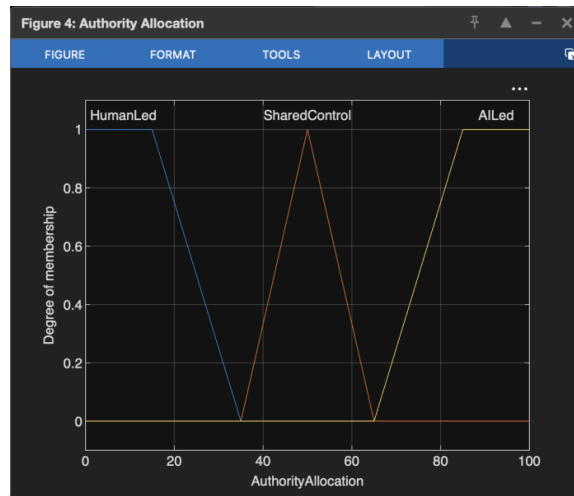


Figure 3.5: Authority Allocation membership functions defining Human-Led, Shared Control and AI-Led decision regions.

Table 3.2 Membership Function Parameters

Variable	Low	Medium	High
Human Readiness	[0 0 20 40]	[30 50 70]	[60 80 100 100]
AI Readiness	[0 0 20 40]	[30 50 70]	[60 80 100 100]
Mission Severity	[0 0 15 35]	[30 50 70]	[65 85 100 100]
Authority Allocation	[0 0 15 35]	[35 50 65]	[65 85 100 100]

3.5 Rule Base Development

The fuzzy rule base represents the knowledge component of the proposed Human–AI Authority Allocation System. The rule base was designed to model expert-inspired reasoning regarding how authority should be allocated between human operators and artificial intelligence systems under varying operational conditions.

Each rule follows a standard fuzzy IF–THEN structure:

IF Human Readiness is X AND AI Readiness is Y AND Mission Severity is Z THEN Authority Allocation is A

where X, Y, Z, and A represent linguistic categories associated with the corresponding variables.

Since each input variable contains three linguistic categories (Low, Medium, and High), the complete rule base contains:

$3 \times 3 \times 3 = 27$ rules

This ensures that every possible combination of input conditions is represented within the system.

The design philosophy of the rule base was guided by three fundamental principles:

1. Higher Human Readiness should generally favour greater human authority.
2. Higher AI Readiness should generally favour greater AI authority.
3. Mission Severity should influence the balance of authority according to operational risk and decision criticality.

For example, when Human Readiness is Low and AI Readiness is High, the system generally favours AI-led decision-making because the AI system is assumed to be better positioned to perform the task. Conversely, when Human Readiness is High and AI Readiness is Low, authority is retained primarily by the human operator.

Situations in which both entities demonstrate comparable readiness frequently result in Shared Control recommendations. This reflects the collaborative nature of Human–AI partnerships and encourages joint decision-making when neither entity clearly dominates.

Table 3.3 Complete Rule Base

Rule	Human Readiness	AI Readiness	Mission Severity	Authority Allocation
R1	Low	Low	Low	Human Led
R2	Low	Low	Medium	Human Led
R3	Low	Low	High	Shared Control
R4	Low	Medium	Low	Shared Control
R5	Low	Medium	Medium	Shared Control
R6	Low	Medium	High	AI Led
R7	Low	High	Low	AI Led
R8	Low	High	Medium	AI Led
R9	Low	High	High	AI Led
R10	Medium	Low	Low	Human Led
R11	Medium	Low	Medium	Shared Control

R12	Medium	Low	High	Shared Control
R13	Medium	Medium	Low	Shared Control
R14	Medium	Medium	Medium	Shared Control
R15	Medium	Medium	High	Shared Control
R16	Medium	High	Low	AI Led
R17	Medium	High	Medium	AI Led
R18	Medium	High	High	AI Led
R19	High	Low	Low	Human Led
R20	High	Low	Medium	Human Led
R21	High	Low	High	Human Led
R22	High	Medium	Low	Human Led
R23	High	Medium	Medium	Shared Control
R24	High	Medium	High	Shared Control
R25	High	High	Low	Shared Control
R26	High	High	Medium	Shared Control
R27	High	High	High	AI Led

The rule base provides complete coverage of the operational space and ensures that every possible combination of inputs generates a valid authority allocation recommendation.

3.5.1 Rule Justification

The fuzzy rule base was developed using expert-inspired reasoning based on three core principles. First, higher Human Readiness should generally favour increased human authority because a capable operator is more likely to make effective decisions. Second, higher AI Readiness should favour increased AI authority because the system demonstrates greater reliability and operational capability. Third, Mission Severity influences the balance of authority by encouraging more cautious decision-making under critical conditions.

Several representative rules are justified below:

- Low Human Readiness + High AI Readiness + High Mission Severity → AI-Led In this situation the AI system is significantly more capable than the human operator and is therefore assigned greater authority.
- High Human Readiness + Low AI Readiness + High Mission Severity → Human-Led The human operator retains authority because the AI system lacks sufficient readiness to manage a critical situation reliably.
- Medium Human Readiness + Medium AI Readiness + High Mission Severity → Shared Control Neither decision-maker clearly dominates, making collaborative decision-making the most appropriate outcome.
- High Human Readiness + High AI Readiness + Medium Mission Severity → Shared Control Both entities demonstrate strong capability and can contribute jointly to decision quality.

The remaining rules follow the same reasoning framework and collectively ensure complete coverage of the operational decision space.

3.6 Fuzzy Inference Mechanism

The proposed system employs a Mamdani Fuzzy Inference System to process the input variables and generate authority allocation recommendations. Mamdani inference was selected because of its interpretability, simplicity, and widespread adoption within decision-support applications (Mamdani and Assilian, 1975).

The inference process consists of four sequential stages:

1. Fuzzification
2. Rule Evaluation
3. Aggregation
4. Defuzzification



Figure 3.6: Mamdani fuzzy inference workflow used to transform crisp operational inputs into authority all

During fuzzification, crisp numerical values are converted into fuzzy membership values. For example, a Human Readiness value of 65 may partially belong to both the Medium and High fuzzy sets simultaneously.

The rule evaluation stage applies the twenty-seven fuzzy rules to determine which rules are activated and to what degree. The activation strength of each rule depends on the membership values of the corresponding input variables.

The system utilises the minimum (MIN) operator for fuzzy conjunctions. Consequently, the activation strength of each rule is determined by the smallest membership value associated with the rule antecedents.

Following rule evaluation, all activated rule outputs are combined through the aggregation process. The maximum (MAX) operator is used to merge the resulting fuzzy output sets into a single aggregated output.

The aggregated fuzzy output is subsequently converted into a crisp authority allocation score through defuzzification.

3.7 Defuzzification Strategy

Defuzzification is the final stage of the fuzzy inference process and is responsible for converting fuzzy output sets into a numerical authority allocation score.

Several defuzzification techniques have been proposed within the literature, including:

- Centroid Method
- Bisector Method
- Mean of Maximum
- Smallest of Maximum
- Largest of Maximum

Among these approaches, the Centroid Method is one of the most widely used because it considers the entire aggregated output distribution when calculating the final result (Ross, 2010).

The centroid method determines the centre of gravity of the aggregated fuzzy output and computes a single numerical value according to:

$$z^* = \frac{\int z \mu_A(z) dz}{\int \mu_A(z) dz}$$

Fig 3.7: where z represents the output variable and $\mu_A(z)$ represents the aggregated fuzzy output membership function.

The proposed system employs centroid defuzzification through MATLAB's built-in fuzzy inference engine. This approach was selected because it generates smooth output transitions and provides stable authority allocation recommendations across the full operating range.

The resulting output value ranges between 0 and 100 and is interpreted according to three authority allocation regions:

- 0–35 → Human Led
- 35–65 → Shared Control
- 65–100 → AI Led

These ranges provide a straightforward interpretation framework that enables users to quickly understand system recommendations.

3.8 Chapter Summary

This chapter presented the design of the proposed Human–AI Authority Allocation System. The system architecture was introduced and the rationale for selecting Human Readiness, AI Readiness, and Mission Severity as input variables was discussed. Membership functions were developed using trapezoidal and triangular fuzzy sets to support gradual transitions between operational states.

A comprehensive rule base consisting of twenty-seven fuzzy rules was constructed to represent authority allocation reasoning across all possible combinations of input conditions. The chapter also described the Mamdani fuzzy inference process and justified the selection of centroid defuzzification for output generation.

The next chapter describes the implementation of the proposed system using MATLAB and the Fuzzy Logic Toolbox, demonstrating how the theoretical design was translated into a fully functional decision-support application.

System Implementation

4.1 Introduction

This chapter describes the implementation of the proposed Human–AI Authority Allocation System using MATLAB and the Fuzzy Logic Toolbox. The implementation translates the theoretical design presented in Chapter 3 into a functional fuzzy inference system capable of generating authority allocation recommendations.

The implementation process consisted of five major stages:

1. Creation of the Mamdani Fuzzy Inference System.
2. Definition of input and output variables.
3. Construction of membership functions.
4. Development of the fuzzy rule base.
5. Scenario testing and performance evaluation.



Figure 4.1: MATLAB implementation workflow followed during development of the Human–AI Authority Allocation System.

4.2 Creation of the Fuzzy Inference System

The first implementation step involved creating a Mamdani Fuzzy Inference System. MATLAB's `mamfis()` function was used to initialise the fuzzy controller and define the inference configuration.

Code Snippet 4.1: Creation and Configuration of the Mamdani Fuzzy Inference System

```
22      %% Create FIS
23
24      fis = mamfis( ...
25          'Name','HumanAIAuthoritySystem', ...
26          'AndMethod','min', ...
27          'OrMethod','max', ...
28          'ImplicationMethod','min', ...
29          'AggregationMethod','max', ...
30          'DefuzzificationMethod','centroid');
```

Figure 4.2: MATLAB implementation of the Mamdani Fuzzy Inference System showing the selected inference operators, aggregation method and centroid defuzzification strategy.

The MIN operator was selected for conjunction operations, while the MAX operator was selected for aggregation. Centroid defuzzification was chosen because it provides smooth and stable output behaviour while considering the complete output distribution.

4.3 Input Variable Implementation

Three input variables were implemented:

- Human Readiness
- AI Readiness
- Mission Severity

Each variable was defined across a numerical range of 0 to 100.

Code Snippet 4.2: Definition of the Human Readiness input variable within the fuzzy inference system.

```
32 %% =====  
33 % HUMAN READINESS  
34 %% =====  
35  
36 fis = addInput(fis,[0 100],'Name','HumanReadiness');  
37  
38 fis = addMF(fis,'HumanReadiness','trapmf',...  
39 [0 0 20 40],'Name','Low');  
40  
41 fis = addMF(fis,'HumanReadiness','trimf',...  
42 [30 50 70],'Name','Medium');  
43  
44 fis = addMF(fis,'HumanReadiness','trapmf',...  
45 [60 80 100 100],'Name','High');
```

Figure 4.3: Creation of the Human Readiness input variable with a numerical range of 0–100.

Code Snippet 4.3: Definition of the AI Readiness input variable.

```
47 %% =====  
48 % AI READINESS  
49 %% =====  
50  
51 fis = addInput(fis,[0 100],'Name','AIReadiness');  
52  
53 fis = addMF(fis,'AIReadiness','trapmf',...  
54 [0 0 20 40],'Name','Low');  
55  
56 fis = addMF(fis,'AIReadiness','trimf',...  
57 [30 50 70],'Name','Medium');  
58  
59 fis = addMF(fis,'AIReadiness','trapmf',...  
60 [60 80 100 100],'Name','High');
```

Figure 4.4: Creation of the AI Readiness input variable used to represent AI operational capability.

Code Snippet 4.4: Definition of the Mission Severity input variable.

```
62 %% =====  
63 % MISSION SEVERITY  
64 %% =====  
65  
66 fis = addInput(fis,[0 100],'Name','MissionSeverity');  
67  
68 fis = addMF(fis,'MissionSeverity','trapmf',...  
69 [0 0 15 35],'Name','Low');  
70  
71 fis = addMF(fis,'MissionSeverity','trimf',...  
72 [30 50 70],'Name','Medium');  
73  
74 fis = addMF(fis,'MissionSeverity','trapmf',...  
75 [65 85 100 100],'Name','High');
```

Figure 4.5: Creation of the Mission Severity input variable used to represent operational criticality.

These variables collectively represent the operational state of the Human–AI system.

4.4 Membership Function Implementation

Membership functions were implemented using a combination of trapezoidal and triangular fuzzy sets.

Trapezoidal functions were used for Low and High categories because they provide stable representation of extreme conditions. Triangular functions were used for Medium categories to facilitate smooth transitions between neighbouring regions.

Code Snippet 4.5: Implementation of Human Readiness membership functions using trapezoidal and triangular fuzzy sets.

```
32 %% =====  
33 % HUMAN READINESS  
34 %% =====  
35  
36 fis = addInput(fis,[0 100],'Name','HumanReadiness');  
37  
38 fis = addMF(fis,'HumanReadiness','trapmf',...  
39 [0 0 20 40],'Name','Low');  
40  
41 fis = addMF(fis,'HumanReadiness','trimf',...  
42 [30 50 70],'Name','Medium');  
43  
44 fis = addMF(fis,'HumanReadiness','trapmf',...  
45 [60 80 100 100],'Name','High');
```

Figure 4.6: MATLAB implementation of Low, Medium and High Human Readiness membership functions.

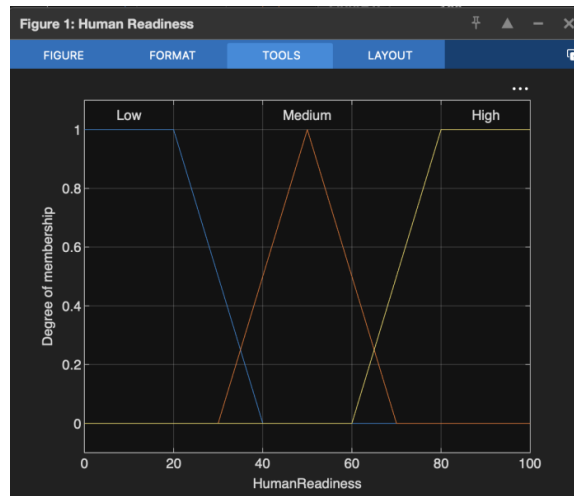


Figure 4.7: Human Readiness membership functions illustrating overlap between Low, Medium and High readiness states.

Code Snippet 4.6: Implementation of AI Readiness membership functions.

```

47  %% =====
48  % AI READINESS
49  %% =====
50
51  fis = addInput(fis,[0 100],'Name','AIReadiness');
52
53  fis = addMF(fis,'AIReadiness','trapmf',...
54  [0 0 20 40],'Name','Low');
55
56  fis = addMF(fis,'AIReadiness','trimf',...
57  [30 50 70],'Name','Medium');
58
59  fis = addMF(fis,'AIReadiness','trapmf',...
60  [60 80 100 100],'Name','High');

```

Figure 4.8: MATLAB implementation of AI Readiness fuzzy sets.

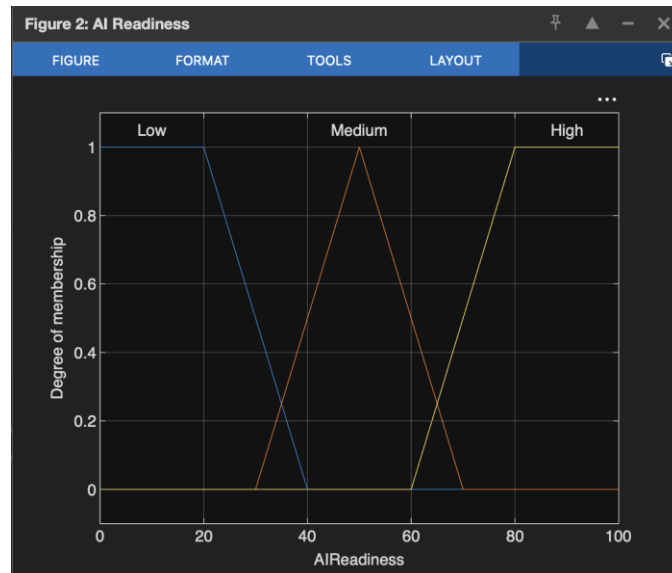


Figure 4.9: AI Readiness membership functions showing gradual transitions between capability levels.

Code Snippet 4.7: Implementation of Mission Severity membership functions.

```

62 %% =====
63 % MISSION SEVERITY
64 %% =====
65
66 fis = addInput(fis,[0 100],'Name','MissionSeverity');
67
68 fis = addMF(fis,'MissionSeverity','trapmf',...
69 [0 0 15 35],'Name','Low');
70
71 fis = addMF(fis,'MissionSeverity','trimf',...
72 [30 50 70],'Name','Medium');
73
74 fis = addMF(fis,'MissionSeverity','trapmf',...
75 [65 85 100 100],'Name','High');

```

Figure 4.10: MATLAB implementation of Mission Severity fuzzy sets.

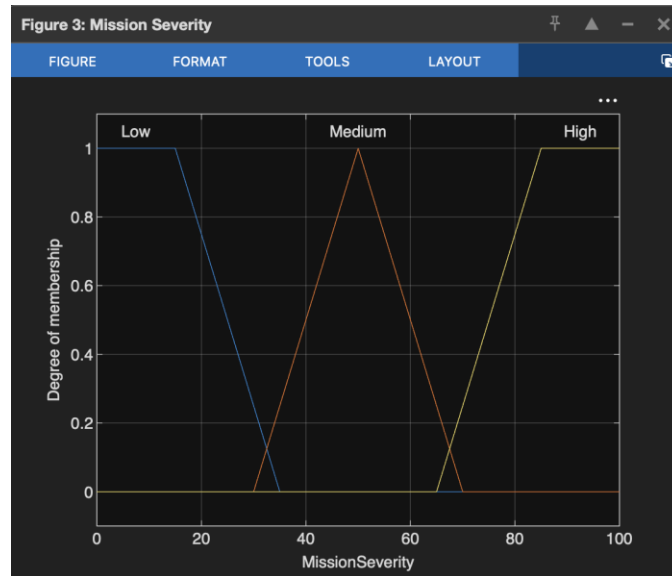


Figure 4.11: Mission Severity membership functions representing low, medium and high operational criticality.

Code Snippet 4.8: Implementation of Authority Allocation output membership functions.

```

77 %% =====
78 % OUTPUT
79 %% =====
80
81 fis = addOutput(fis,[0 100],'Name','AuthorityAllocation');
82
83 fis = addMF(fis,'AuthorityAllocation','trapmf',...
84 [0 0 15 35],'Name','HumanLed');
85
86 fis = addMF(fis,'AuthorityAllocation','trimf',...
87 [35 50 65],'Name','SharedControl');
88
89 fis = addMF(fis,'AuthorityAllocation','trapmf',...
90 [65 85 100 100],'Name','AILED');

```

Figure 4.12: MATLAB implementation of output membership functions.

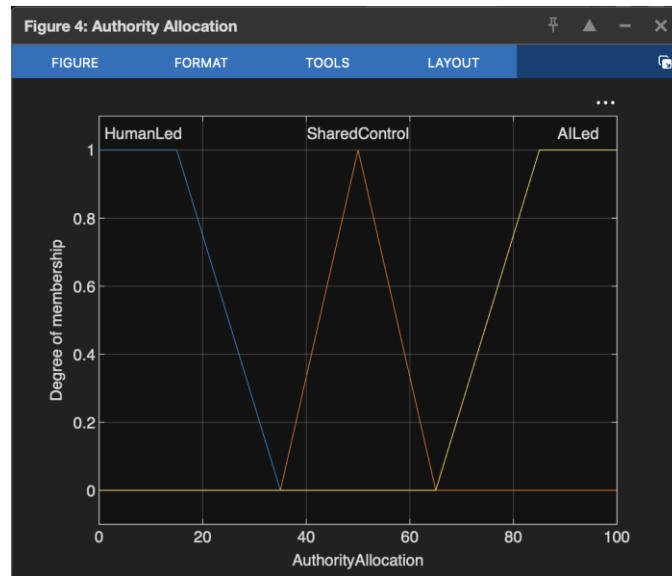


Figure 4.13: Authority Allocation membership functions defining Human-Led, Shared Control and AI-Led decision regions.

Table 4.1 Membership Function Parameters

Variable	Low	Medium	High
Human Readiness	[0 0 20 40]	[30 50 70]	[60 80 100 100]
AI Readiness	[0 0 20 40]	[30 50 70]	[60 80 100 100]
Mission Severity	[0 0 15 35]	[30 50 70]	[65 85 100 100]
Authority Allocation	[0 0 15 35]	[35 50 65]	[65 85 100 100]

4.5 Rule Base Implementation

The fuzzy rule base was implemented using MATLAB's `addRule()` function. A total of twenty-seven rules were defined to cover all possible combinations of Human Readiness, AI Readiness, and Mission Severity.

Code Snippet 4.9: Definition of the 27-rule fuzzy knowledge base used for authority allocation reasoning.

```

106 ruleList = [
107
108 % HUMAN LOW
109 1 1 1 1 1
110 1 1 2 1 1
111 1 1 3 2 1
112
113 1 2 1 2 1
114 1 2 2 2 1
115 1 2 3 3 1
116 |
117 1 3 1 3 1
118 1 3 2 3 1
119 1 3 3 3 1
120
121 % HUMAN MEDIUM
122 2 1 1 1 1
123 2 1 2 2 1
124 2 1 3 2 1
125
126 2 2 1 2 1
127 2 2 2 2 1
128 2 2 3 2 1
129
130 2 3 1 3 1
131 2 3 2 3 1
132 2 3 3 3 1
133
134 % HUMAN HIGH
135 3 1 1 1 1
136 3 1 2 1 1
137 3 1 3 1 1
138
139 3 2 1 1 1
140 3 2 2 2 1
141 3 2 3 2 1
142
143 3 3 1 2 1
144 3 3 2 2 1
145 3 3 3 3 1
146
147 ];
148
149 fis = addRule(fis,ruleList);

```

Figure 4.14: Partial MATLAB implementation of the fuzzy rule base.

The twenty-seven fuzzy rules were implemented using MATLAB's `addRule()` function. The complete rule base is presented in Chapter 3.

4.6 Rule Viewer Implementation

To improve transparency and support system verification, MATLAB's Rule Inference Viewer was used to validate and visualise the behaviour of the fuzzy inference system. The interface displays activated rules, membership function responses, and the resulting authority allocation recommendation for selected input values.

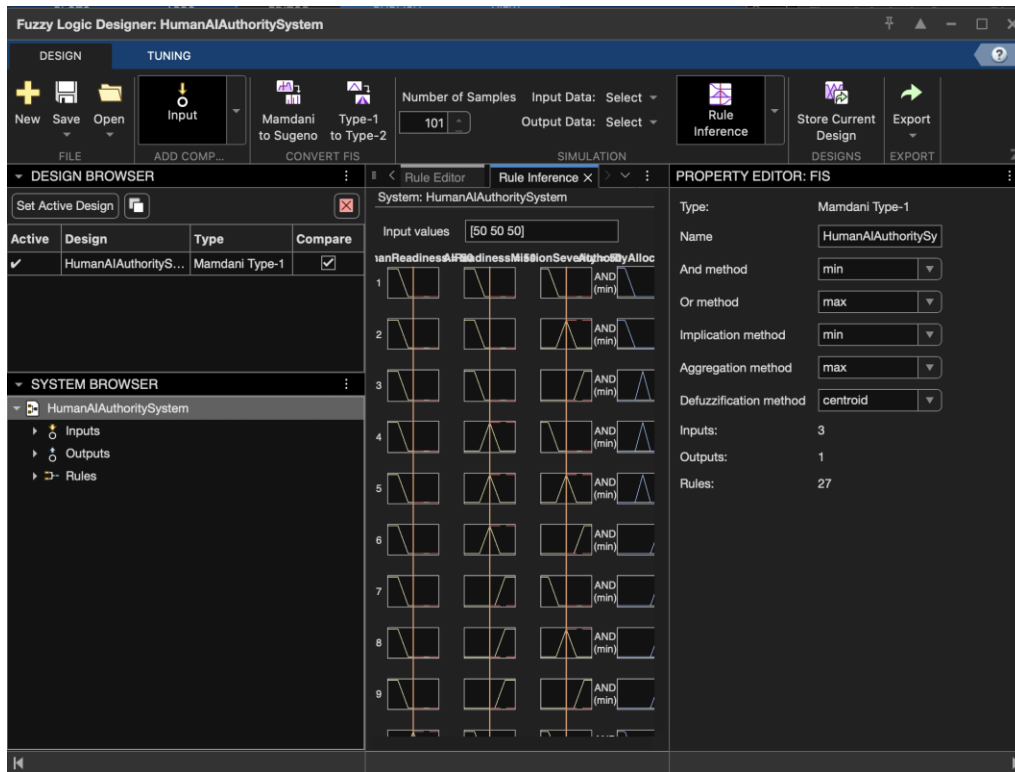


Figure 4.15: MATLAB Rule Inference Viewer displaying activated fuzzy rules, membership values, and authority allocation inference for the Human–AI Authority Allocation System.

4.7 Scenario Testing Implementation

Following implementation, the fuzzy inference system was evaluated using a range of testing scenarios. The objective of testing was to verify that the system generated reasonable authority allocation recommendations across different operational conditions.

A total of thirty test cases were used. These consisted of:

- Original operational scenarios
- Boundary condition scenarios
- Additional evaluation scenarios

Code Snippet 4.10: Definition of the scenario testing dataset used for system evaluation.

```

192     testData = [
193
194         % Original Scenarios
195         20 30 80
196         20 90 90
197         90 20 20
198         60 90 95
199         70 70 50
200         15 95 85
201         95 15 40
202         50 50 50
203         40 85 95
204         90 20 50
205
206         % Boundary Cases
207         0 0 0
208         0 0 100
209         0 100 0
210         0 100 100
211
212         100 0 0
213         100 0 100
214         100 100 0
215         100 100 100
216
217         % Additional Evaluation Cases
218         25 25 25
219         25 25 75
220         25 75 25
221         25 75 75
222
223         75 25 25
224         75 25 75
225         75 75 25
226         75 75 75
227
228         50 20 80
229         50 80 20
230         20 50 80
231         80 50 20
232
233     ];

```

Figure 4.16: MATLAB implementation of the test scenario dataset containing operational, boundary and evaluation cases.

The dataset includes operational, boundary, and evaluation scenarios to assess system behaviour under different conditions.

Table 4.2 Complete Testing Dataset

Case	Human	AI	Severity
1	20	30	80
2	20	90	90
3	90	20	20
4	60	90	95
5	70	70	50
6	15	95	85
7	95	15	40
8	50	50	50

9	40	85	95
10	90	20	50
11	0	0	0
12	0	0	100
13	0	100	0
14	0	100	100
15	100	0	0
16	100	0	100
17	100	100	0
18	100	100	100
19	25	25	25
20	25	25	75
21	25	75	25
22	25	75	75
23	75	25	25
24	75	25	75
25	75	75	25
26	75	75	75
27	50	20	80
28	50	80	20
29	20	50	80
30	80	50	20

4.8 Evaluation and Classification Implementation

The authority allocation score generated by the fuzzy inference system was converted into three decision categories:

- Human Led
- Shared Control
- AI Led

This classification process improves interpretability by translating numerical outputs into operational recommendations.

Code Snippet 4.11: Classification logic used to convert numerical authority scores into operational authority categories.

```

262 AuthorityCategory = strings(length(results),1);
263
264 for i = 1:length(results)
265     if results(i) < 35
266         AuthorityCategory(i) = "Human Led";
267     elseif results(i) < 65
268         AuthorityCategory(i) = "Shared Control";
269     else
270         AuthorityCategory(i) = "AI Led";
271     end
272 end
273
274
275
276

```

Figure 4.17: MATLAB implementation of authority allocation categorisation into Human-Led, Shared Control and AI-Led outcomes.

The classification thresholds were defined as:

Table 4.3: Mapping of authority allocation scores to operational decision categories.

Authority Score	Classification
0 – 35	Human Led
35 – 65	Shared Control
65 – 100	AI Led

This categorisation framework enables rapid interpretation of authority allocation recommendations.

4.9 Results Table Generation

A structured results table was generated to store all scenario inputs, authority allocation scores, and associated authority categories.

Code Snippet 4.12: Construction of the results table for storing scenario inputs, authority scores and classifications.

```

278 %% =====
279 % CREATE RESULTS TABLE
280 %% =====
281
282 ResultsTable = table( ...
283     ScenarioNames,...
284     testData(:,1),...
285     testData(:,2),...
286     testData(:,3),...
287     results,...
288     AuthorityCategory,...
289     'VariableNames',...
290     {'Scenario',...
291      'HumanReadiness',...
292      'AIReadiness',...
293      'MissionSeverity',...
294      'AuthorityAllocation',...
295      'AuthorityCategory'});
296
297 disp(ResultsTable)

```

Figure 4.18: MATLAB implementation of the structured results table.

The generated table forms the primary dataset used during performance evaluation and reporting.

Scenario	HumanReadiness	AIReadiness	MissionSeverity	AuthorityAllocation	AuthorityCategory
"Case_1"	20	30	80	50	"Shared Control"
"Case_2"	20	90	90	87.098	"AI Led"
"Case_3"	90	20	20	13.833	"Human Led"
"Case_4"	60	90	95	85.115	"AI Led"
"Case_5"	70	70	50	50	"Shared Control"
"Case_6"	15	95	85	87.098	"AI Led"
"Case_7"	95	15	40	14.885	"Human Led"
"Case_8"	50	50	50	50	"Shared Control"
"Case_9"	40	85	95	85.115	"AI Led"
"Case_10"	90	20	50	12.902	"Human Led"
"Case_11"	0	0	0	12.902	"Human Led"
"Case_12"	0	0	100	50	"Shared Control"
"Case_13"	0	100	0	87.098	"AI Led"
"Case_14"	0	100	100	87.098	"AI Led"
"Case_15"	100	0	0	12.902	"Human Led"
"Case_16"	100	0	100	12.902	"Human Led"
"Case_17"	100	100	0	50	"Shared Control"
"Case_18"	100	100	100	87.098	"AI Led"
"Case_19"	25	25	25	14.885	"Human Led"
"Case_20"	25	25	75	50	"Shared Control"
"Case_21"	25	75	25	85.115	"AI Led"
"Case_22"	25	75	75	85.115	"AI Led"
"Case_23"	75	25	25	14.885	"Human Led"
"Case_24"	75	25	75	14.885	"Human Led"
"Case_25"	75	75	25	50	"Shared Control"
"Case_26"	75	75	75	85.115	"AI Led"
"Case_27"	50	20	80	50	"Shared Control"
"Case_28"	50	80	20	86.167	"AI Led"
"Case_29"	20	50	80	86.167	"AI Led"
"Case_30"	80	50	20	13.833	"Human Led"

Figure 4.19: Generated scenario evaluation results showing authority allocation scores and corresponding authority categories.

4.10 Performance Metrics Implementation

Several descriptive performance metrics were calculated to provide a high-level summary of system behaviour.

Code Snippet 4.13: Calculation of descriptive performance metrics for system evaluation.

```
249 MeanAuthority = mean(results);  
250 MaxAuthority = max(results);  
251 MinAuthority = min(results);  
252 StdAuthority = std(results);
```

Figure 4.20: MATLAB implementation of authority allocation performance metrics.

The calculated metrics include:

- Mean Authority Allocation
- Maximum Authority Allocation
- Minimum Authority Allocation
- Standard Deviation

These measures provide insight into the overall distribution and variability of authority allocation recommendations.

```
===== PERFORMANCE METRICS =====  
Mean Authority Allocation = 52.41  
Maximum Authority Allocation = 87.10  
Minimum Authority Allocation = 12.90  
Standard Deviation = 31.37
```

Figure 4.21: Statistical summary of authority allocation outputs generated by the fuzzy inference system.

4.11 Results Export

To support reproducibility and further analysis, the implementation automatically exports generated outputs to external files.

Code Snippet 4.14: Export of evaluation results to CSV and Excel files for reproducibility and further analysis.

```
310 %% =====
311 % EXPORT CSV
312 %% =====
313
314 writetable(ResultsTable,...
315     'AuthorityAllocationResults.csv');
316
317 writetable(ResultsTable,...
318     'Table1_TestResults.xlsx');
319
320 MetricsTable = table( ...
321     MeanAuthority,...
322     MaxAuthority,...
323     MinAuthority,...
324     StdAuthority);
325
326 writetable(MetricsTable,...
327     'PerformanceMetrics.csv');
```

Figure 4.22: MATLAB implementation of automated results export functionality.

The exported files include:

- AuthorityAllocationResults.csv
- Table1_TestResults.xlsx
- PerformanceMetrics.csv

These outputs provide a permanent record of evaluation results and facilitate future analysis.

4.12 Visualisation Implementation

Two graphical visualisations were developed to assist interpretation of system outputs.

The first visualisation displays authority allocation scores across all test scenarios.

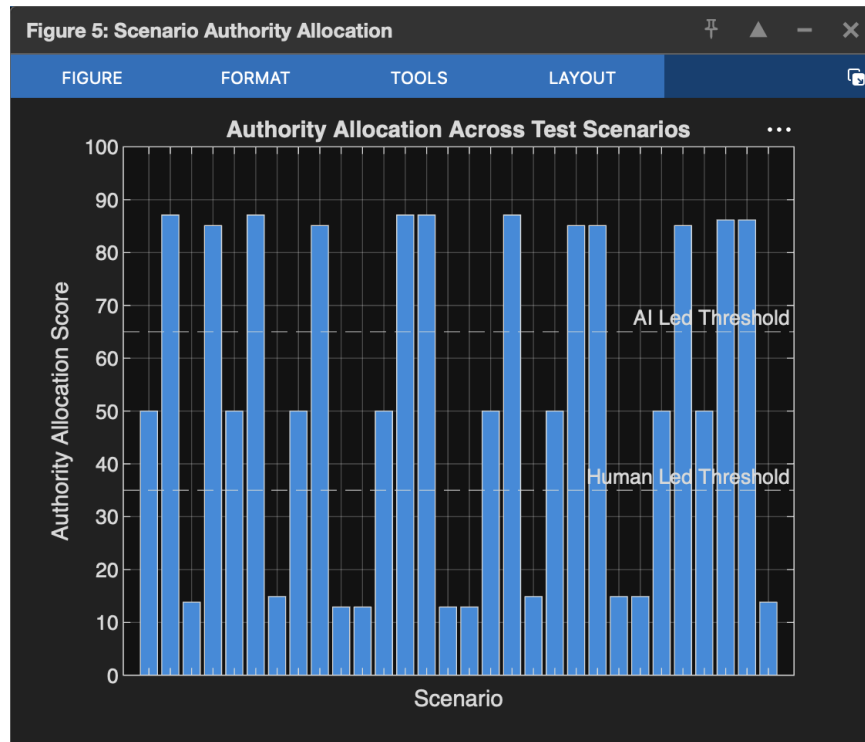


Figure 4.23: Authority allocation scores generated across thirty test scenarios.

The second visualisation summarises the number of Human-Led, Shared Control, and AI-Led recommendations generated during testing.

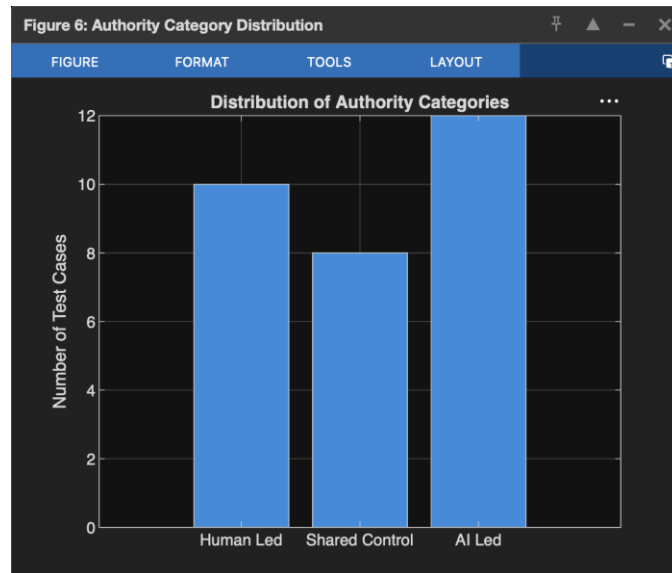


Figure 4.24: Distribution of Human-Led, Shared Control and AI-Led recommendations generated during evaluation.

Together, these visualisations provide an intuitive overview of system behaviour and support evaluation of authority allocation patterns.

4.13 Chapter Summary

This chapter described the implementation of the proposed Human–AI Authority Allocation System using MATLAB and the Fuzzy Logic Toolbox. The implementation included creation of a Mamdani fuzzy inference system, development of membership functions, construction of a

twenty-seven-rule knowledge base, scenario testing, performance evaluation, and automated result export.

The completed implementation successfully transformed the theoretical design presented in Chapter 3 into a functional decision-support system capable of generating interpretable authority allocation recommendations.

The next chapter presents the experimental results and evaluation of the proposed system, including analysis of membership functions, authority allocation behaviour, testing outcomes, and system performance.

Results and Evaluation

5.1 Introduction

This chapter presents the results obtained from testing the proposed Human–AI Authority Allocation System. The objective of the evaluation was to determine whether the fuzzy inference system produced logical authority allocation recommendations under a variety of operational conditions.

The evaluation focuses on four areas:

1. Verification of membership function behaviour.
2. Analysis of authority allocation outputs.
3. Assessment of testing results across all scenarios.
4. Evaluation of overall system performance.

The results generated during implementation are analysed and interpreted to determine whether the proposed fuzzy inference system successfully meets the project objectives.

5.2 Verification of Membership Functions

Membership functions form the foundation of the fuzzy inference process. Therefore, the first stage of evaluation involved verifying that all membership functions correctly represented the intended linguistic concepts.

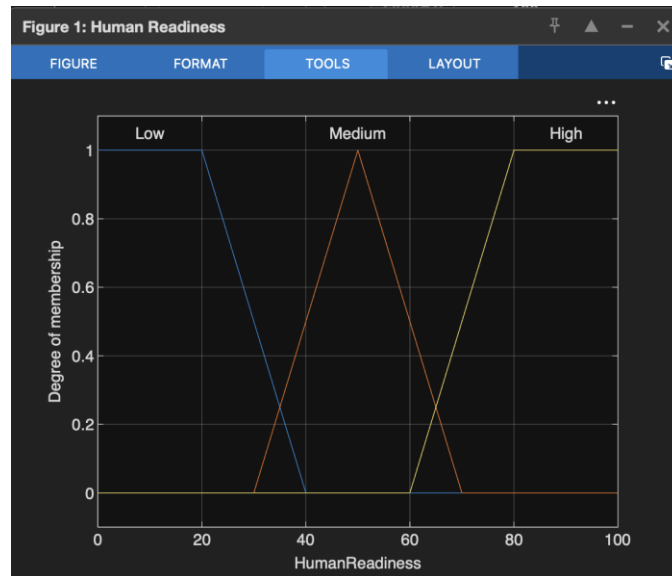


Figure 5.1: Human Readiness membership functions illustrating overlap between Low, Medium and High readiness states.

The figure demonstrates appropriate overlap between the Low, Medium, and High fuzzy sets. This overlap ensures smooth transitions between readiness levels and prevents abrupt changes in authority allocation recommendations.

Similarly, the AI Readiness membership functions shown in Figure provide consistent representation of AI capability levels.

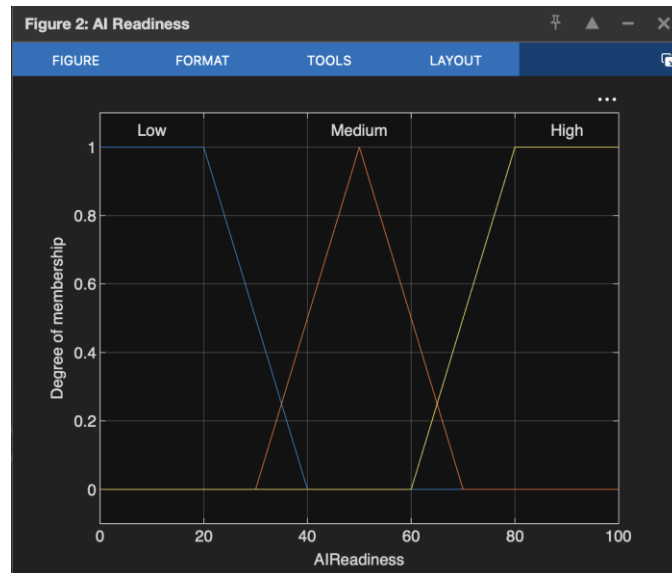


Figure 5.2: AI Readiness membership functions showing gradual transitions between capability levels.

The symmetrical structure used for Human Readiness and AI Readiness ensures balanced treatment of both decision-making entities within the fuzzy inference process.

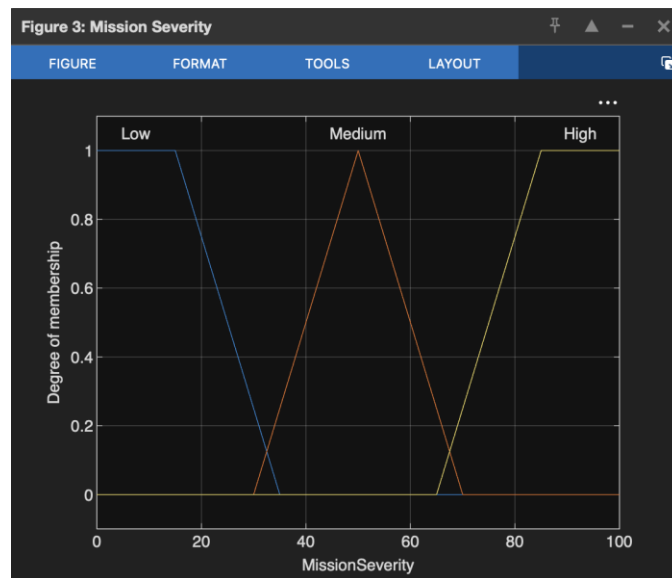


Figure 5.3: Mission Severity membership functions representing low, medium and high operational criticality.

The design ensures that low-risk and high-risk situations are clearly distinguishable while still allowing moderate-risk situations to partially belong to multiple categories when appropriate.

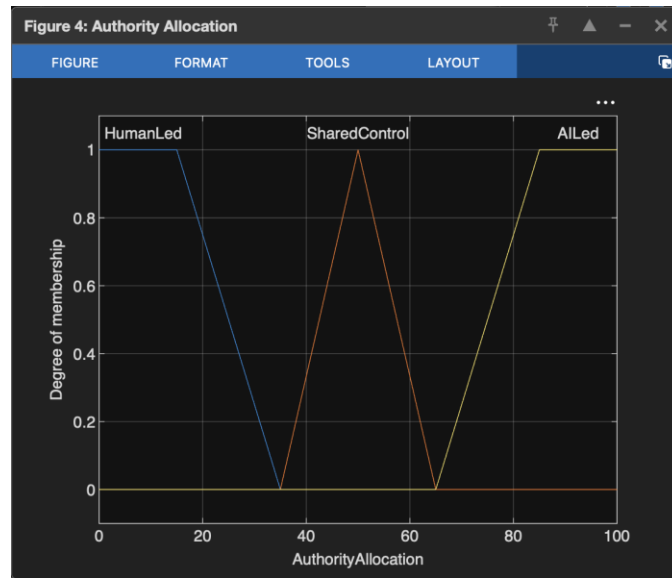


Figure 5.4: Authority Allocation membership functions defining Human-Led, Shared Control and AI-Led decision regions.

The output structure provides three clearly interpretable authority allocation categories:

- Human Led
- Shared Control
- AI Led

Collectively, the membership functions provide smooth transitions across the complete operating range and support realistic authority allocation recommendations.

5.3 Rule Base Evaluation

The effectiveness of the fuzzy inference system depends largely on the quality of the rule base.

The proposed system contains twenty-seven rules representing all possible combinations of Human Readiness, AI Readiness, and Mission Severity.

The complete rule base presented in Chapter 3 was evaluated to verify consistency and coverage across all possible input combinations.

Inspection of the rule base confirms that all input combinations are represented and that no operational conditions remain uncovered.

The rules exhibit consistent behaviour aligned with the design objectives. Higher levels of AI readiness generally increase authority delegation to the AI system, while higher levels of human readiness increase human involvement in decision-making.

Shared Control recommendations occur primarily when both entities demonstrate comparable capability levels. This behaviour reflects the collaborative philosophy underpinning Human–AI partnerships.

The rule base therefore provides complete and logically consistent coverage of the authority allocation problem space.

5.4 Scenario Testing Results

The fuzzy inference system was evaluated using thirty test cases designed to represent a variety of operational conditions.

The fuzzy inference system was evaluated using thirty test scenarios previously presented in Table 4.2. The testing dataset consisted of operational scenarios, boundary-condition cases, and additional evaluation scenarios designed to assess system behaviour across a broad range of Human Readiness, AI Readiness, and Mission Severity combinations. The results obtained from these scenarios are analysed and discussed in the following sections.

The results demonstrate that the system responds appropriately to changing operational conditions.

Several examples illustrate the behaviour of the fuzzy inference system:

Case 2

Human Readiness = 20

AI Readiness = 90

Mission Severity = 90

The resulting authority allocation recommendation falls within the AI-Led region.

This outcome is logical because the AI system demonstrates significantly greater capability than the human operator while operating within a highly critical environment.

Case 7

Human Readiness = 95

AI Readiness = 15

Mission Severity = 40

The system recommends Human-Led control.

This recommendation reflects the substantially greater capability of the human operator relative to the AI system.

Case 8

Human Readiness = 50

AI Readiness = 50

Mission Severity = 50

The system recommends Shared Control.

This outcome demonstrates that the fuzzy inference system appropriately recognises balanced capability levels and allocates authority collaboratively. The result is particularly important because it indicates that the model does not favour either decision-making entity when their readiness levels are similar. Instead, authority is shared between both parties, which aligns with the intended objective of Human–AI cooperation.

These examples indicate that the system produces recommendations consistent with expected operational reasoning.

5.4.1 Sensitivity Analysis

A sensitivity analysis was conducted to evaluate how changes in input values influence authority allocation recommendations. Human Readiness and Mission Severity were fixed at 50, while AI Readiness was varied across its operating range.

As AI Readiness increased from low to high values, the authority allocation score increased progressively towards the AI-Led region. This behaviour indicates that the fuzzy inference

system responds smoothly to changes in operational conditions rather than producing abrupt transitions.

The observed trend confirms that the membership functions and rule base interact appropriately to support gradual authority allocation decisions. Such behaviour is desirable because real-world Human–AI collaboration environments rarely experience sudden changes in responsibility. Instead, authority should transition progressively as confidence in either decision-making entity changes.

5.4.2 Boundary Case Validation

The system was tested using extreme boundary conditions to verify logical behaviour. When Human Readiness and AI Readiness were both assigned values of 0, the system produced conservative authority allocation recommendations. Conversely, scenarios involving maximum AI Readiness and low Human Readiness resulted in AI-Led recommendations. Cases where Human Readiness exceeded AI Readiness produced Human-Led outcomes. These results indicate that the fuzzy rule base behaves consistently under extreme operating conditions and maintains logical authority allocation behaviour across the full input range.

5.5 Authority Allocation Distribution

To better understand overall system behaviour, the authority allocation recommendations were grouped into three categories:

- Human Led
- Shared Control
- AI Led

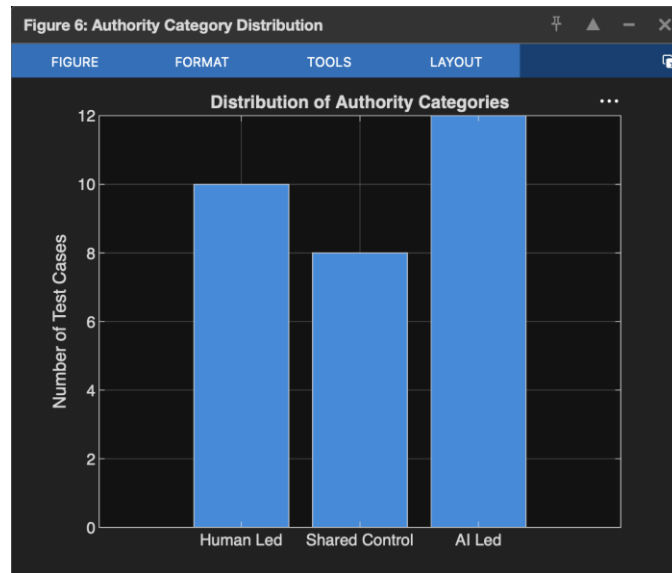


Figure 5.5: Distribution of authority allocation recommendations across Human-Led, Shared Control, and AI-Led categories.

The figure indicates that all three authority categories were utilised during testing.

This result is important because it demonstrates that the system does not exhibit bias toward a single authority allocation outcome. Instead, recommendations vary according to the operational conditions presented to the fuzzy inference system.

The presence of all three categories confirms that the rule base successfully differentiates between diverse operating conditions.

5.6 Authority Allocation Trends

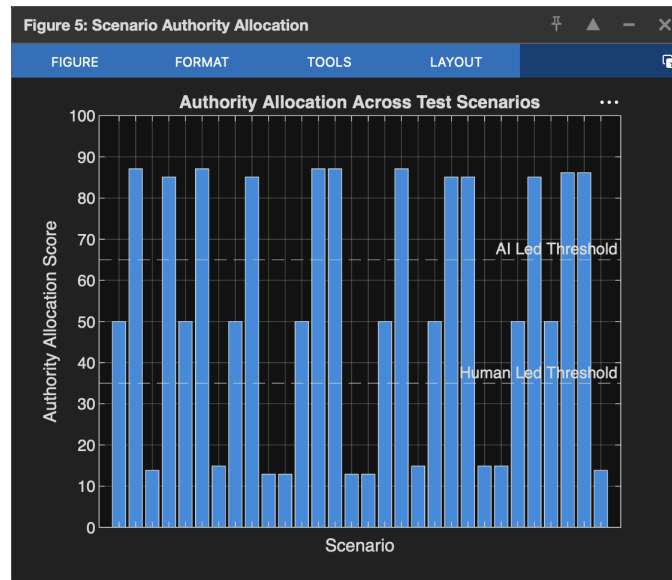


Figure 5.6: Variation in authority allocation recommendations under different operational conditions.

The chart demonstrates significant variation across scenarios, indicating that the fuzzy inference system is sensitive to changes in Human Readiness, AI Readiness, and Mission Severity.

Authority allocation scores span a broad range of the output space, demonstrating that the system effectively utilises the available decision range rather than clustering around a narrow set of values.

This behaviour suggests that the membership functions and rule base collectively provide sufficient granularity to distinguish between different operational situations.

5.7 Performance Evaluation

Several statistical metrics were calculated to evaluate overall system behaviour.

Scenario	HumanReadiness	AIReadiness	MissionSeverity	AuthorityAllocation	AuthorityCategory
"Case_1"	20	30	80	50	"Shared Control"
"Case_2"	20	90	90	87.098	"AI Led"
"Case_3"	90	20	20	13.833	"Human Led"
"Case_4"	60	90	95	85.115	"AI Led"
"Case_5"	70	70	50	50	"Shared Control"
"Case_6"	15	95	85	87.098	"AI Led"
"Case_7"	95	15	40	14.885	"Human Led"
"Case_8"	50	50	50	50	"Shared Control"
"Case_9"	40	85	95	85.115	"AI Led"
"Case_10"	90	20	50	12.902	"Human Led"
"Case_11"	0	0	0	12.902	"Human Led"
"Case_12"	0	0	100	50	"Shared Control"
"Case_13"	0	100	0	87.098	"AI Led"
"Case_14"	0	100	100	87.098	"AI Led"
"Case_15"	100	0	0	12.902	"Human Led"
"Case_16"	100	0	100	12.902	"Human Led"
"Case_17"	100	100	0	50	"Shared Control"
"Case_18"	100	100	100	87.098	"AI Led"
"Case_19"	25	25	25	14.885	"Human Led"
"Case_20"	25	25	75	50	"Shared Control"
"Case_21"	25	75	25	85.115	"AI Led"
"Case_22"	25	75	75	85.115	"AI Led"
"Case_23"	75	25	25	14.885	"Human Led"
"Case_24"	75	25	75	14.885	"Human Led"
"Case_25"	75	75	25	50	"Shared Control"
"Case_26"	75	75	75	85.115	"AI Led"
"Case_27"	50	20	80	50	"Shared Control"
"Case_28"	50	80	20	86.167	"AI Led"
"Case_29"	20	50	80	86.167	"AI Led"
"Case_30"	80	50	20	13.833	"Human Led"

Figure 5.7: Statistical summary of authority allocation outputs generated across the testing scenarios.

The mean authority allocation score provides an indication of the overall balance between human authority and AI authority across all test scenarios.

The maximum and minimum values demonstrate that the system successfully produces recommendations across the full authority allocation spectrum.

The standard deviation reflects variability within the output distribution. A moderate standard deviation indicates that the system generates diverse recommendations rather than repeatedly producing similar outputs.

These metrics collectively suggest that the fuzzy inference system responds dynamically to changing operational conditions.

5.8 Strengths of the Proposed System

Several strengths were identified during evaluation.

First, the system provides transparent decision-making through the use of interpretable fuzzy rules. Unlike many machine learning approaches, every authority allocation recommendation can be traced directly to specific rule activations.

Second, the system effectively handles uncertainty through fuzzy membership functions and linguistic reasoning.

Third, the implementation is computationally lightweight and easily understandable, making it suitable for educational and decision-support applications.

Finally, the system successfully integrates human capability, AI capability, and mission criticality into a single decision-making framework.

5.9 Limitations

Although the proposed system performed successfully during testing, several limitations should be acknowledged.

First, the fuzzy rule base was developed using expert-inspired reasoning rather than formally elicited domain expertise. Alternative experts may define different authority allocation strategies, potentially resulting in different system behaviour.

Second, all input variables are treated with equal importance. In real-world Human–AI systems, certain variables such as Mission Severity may have greater influence on authority allocation decisions. Future work could incorporate weighted rules to better reflect operational priorities.

Third, membership function parameters were selected manually based on engineering judgement. Although the chosen parameters produced logical outputs, optimisation techniques could be used to identify more effective parameter configurations.

Fourth, evaluation was conducted using simulated scenarios rather than real-world operational datasets. Consequently, the practical effectiveness of the proposed system remains to be validated in domains such as healthcare, autonomous systems, aviation, and industrial automation.

Finally, the current system is static and does not adapt over time. Future adaptive fuzzy systems could automatically update membership functions and rule structures based on historical performance data.

Despite these limitations, the proposed system successfully demonstrates the suitability of fuzzy logic for Human–AI authority allocation and provides a strong foundation for future development.

5.10 Chapter Summary

This chapter presented the results and evaluation of the proposed Human–AI Authority Allocation System. Membership function verification confirmed appropriate fuzzy set design, while scenario testing demonstrated logical authority allocation behaviour across diverse operating conditions.

The rule base provided complete coverage of the problem space and generated interpretable recommendations consistent with expected operational reasoning. Statistical evaluation and visual analysis further demonstrated that the system responds dynamically to changing input conditions.

Overall, the evaluation indicates that the proposed fuzzy inference system successfully achieves its objective of allocating authority between human operators and AI systems under uncertain conditions.

Conclusion and Future Work

6.1 Conclusion

The objective of this project was to develop a Human–AI Authority Allocation System capable of supporting decision-making under uncertainty through the use of fuzzy logic. Traditional approaches to authority allocation often rely on rigid thresholds or fixed levels of automation that

may not adequately represent the complex and dynamic nature of Human–AI collaboration. To address this challenge, a Mamdani Fuzzy Inference System was designed and implemented using MATLAB and the Fuzzy Logic Toolbox.

The proposed system utilised three input variables: Human Readiness, AI Readiness, and Mission Severity. These variables were selected because they represent key factors influencing authority allocation decisions within collaborative operational environments. Through the use of trapezoidal and triangular membership functions, numerical inputs were transformed into linguistic variables that more closely resemble human reasoning processes.

A comprehensive fuzzy rule base consisting of twenty-seven rules was developed to represent authority allocation knowledge. The rule base enabled the system to evaluate all possible combinations of input conditions and generate interpretable authority allocation recommendations. Centroid defuzzification was used to convert fuzzy outputs into crisp numerical values ranging from 0 to 100.

The implemented system was evaluated using thirty test scenarios covering operational cases, boundary conditions, and additional evaluation situations. Results demonstrated that the system produced logical recommendations that aligned with expected decision-making behaviour. Human-Led recommendations were generated when human capability significantly exceeded AI capability, AI-Led recommendations were generated when AI readiness dominated, and Shared Control recommendations emerged when both entities demonstrated comparable capability levels.

The findings indicate that fuzzy logic provides an effective mechanism for modelling authority allocation decisions under uncertainty. The proposed system successfully achieved the project objectives by providing an interpretable, flexible, and computationally efficient framework for Human–AI authority allocation.

Overall, the project demonstrates the suitability of Mamdani fuzzy inference systems for supporting adaptive decision-making within Human–AI collaborative environments.

6.2 Future Work

Although the proposed system achieved its objectives, several opportunities exist for future enhancement.

First, future implementations could incorporate additional input variables such as operator workload, trust level, environmental uncertainty, or system confidence. Including these factors may improve decision accuracy and provide a more comprehensive representation of Human–AI collaboration.

Second, the current rule base was developed using expert-inspired reasoning. Future studies could involve consultation with domain experts to validate and refine rule definitions. Expert validation would strengthen the practical relevance of the authority allocation recommendations.

Third, adaptive fuzzy systems could be explored. Instead of using fixed membership functions, machine learning techniques could be employed to automatically adjust membership parameters based on historical operational data. Such an approach would enable the system to learn from experience and improve performance over time.

Fourth, future research could investigate hybrid intelligent systems that combine fuzzy logic with machine learning methods such as neural networks or evolutionary optimisation algorithms. These approaches may enhance predictive capability while retaining the interpretability advantages of fuzzy reasoning.

Finally, validation using real-world datasets would provide stronger evidence of practical applicability. Testing the system within aviation, autonomous vehicles, healthcare, or industrial automation environments would enable assessment under realistic operational conditions.

These enhancements represent promising directions for extending the functionality and practical value of Human–AI authority allocation systems.

References

Endsley, M.R. (2017) ‘From Here to Autonomy: Lessons Learned from Human–Automation Research’, *Human Factors*, 59(1), pp. 5–27.

Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. Cambridge, MA: MIT Press.

Hancock, P.A., Billings, D.R., Schaefer, K.E., Chen, J.Y.C., de Visser, E.J. and Parasuraman, R. (2011) ‘A Meta-Analysis of Factors Affecting Trust in Human–Robot Interaction’, *Human Factors*, 53(5), pp. 517–527.

- Kaber, D.B. and Endsley, M.R. (2004) 'The Effects of Level of Automation and Adaptive Automation on Human Performance, Situation Awareness and Workload in a Dynamic Control Task', *Theoretical Issues in Ergonomics Science*, 5(2), pp. 113–153.
- Kosko, B. (1994) *Fuzzy Thinking: The New Science of Fuzzy Logic*. New York: Hyperion.
- Lee, J.D. and See, K.A. (2004) 'Trust in Automation: Designing for Appropriate Reliance', *Human Factors*, 46(1), pp. 50–80.
- Mamdani, E.H. and Assilian, S. (1975) 'An Experiment in Linguistic Synthesis with a Fuzzy Logic Controller', *International Journal of Man–Machine Studies*, 7(1), pp. 1–13.
- Mendel, J.M. (1995) 'Fuzzy Logic Systems for Engineering: A Tutorial', *Proceedings of the IEEE*, 83(3), pp. 345–377.
- Parasuraman, R., Sheridan, T.B. and Wickens, C.D. (2000) 'A Model for Types and Levels of Human Interaction with Automation', *IEEE Transactions on Systems, Man and Cybernetics Part A*, 30(3), pp. 286–297.
- Ross, T.J. (2010) *Fuzzy Logic with Engineering Applications*. 3rd edn. Chichester: John Wiley & Sons.
- Shneiderman, B. (2022) *Human-Centered AI*. New York: Oxford University Press.
- Turban, E., Sharda, R. and Delen, D. (2011) *Decision Support and Business Intelligence Systems*. 9th edn. Upper Saddle River: Pearson.
- Zadeh, L.A. (1965) 'Fuzzy Sets', *Information and Control*, 8(3), pp. 338–353.
- Zimmermann, H.J. (2010) *Fuzzy Set Theory and Its Applications*. 4th edn. Dordrecht: Springer.